Masters of Engineering in ‘Product Design and Development’

Department of Industrial Engineering

University of Engineering and Technology

Taxila

Masters of Engineering in Product Design and Development

Objectives

The Department of Industrial Engineering, University of Engineering and Technology, Taxila, likely offers degree in Master of Engineering in ‘Product Design and Development’ in collaboration with HMC. The MSc courses provide students with a knowledge and understanding of the 'state of the art' in one or more of the many areas of Industrial engineering specifically from product design and development perspective. This combines with the potential for students to develop their abilities in subjects, which are useful in all areas of Industrial engineering wholly and product development in specific and are associated with the application of product design/development in broader range of engineering practice.

Upon completion, graduates of the program will be qualified to work in a multitude of different areas of industry, including research and development, design, manufacturing and production.

Program Outline

This program is run under the post graduate regulations of University of Engineering and Technology, Taxila and existing rules apply. The Master’s Program studying of 24-credit hrs-graduate level courses and 6-research credits hrs for a thesis, with total of 30 credits.

Program Courses

The program includes the following courses.

1. R&D strategy and organization (3 Credits)
2. Product Design (3 Credits)
3. Product Development (3 Credits)
4. Advanced prototyping (3 Credits)
5. Advanced Manufacturing Processes (3 Credits)
6. Product Quality and Assurance (3 Credits)
7. Product Standardization (3 Credits)
8. Marketing and product commercialization (3 Credits)
9. Research Thesis (6 Credits)

COURSE DESCRIPTION:

i. R&D strategy and organization (3 Credits)

Contents

Organizational innovativeness and idea creation, product development process tools, scoping product developments: technical and business concerns, understanding customer needs, establishing product functions, product teardown and experimentation, benchmarking and establishing engineering specifications, product portfolios and portfolio architecture, product architecture, generating concepts, concept selection, concept embodiment, co-ordination and effective organizational structures for product innovation, R&D partnerships and alliances: strategies and processes, transferring knowledge, innovation culture and managing professionals.

Objectives

- Understand R&D from a strategic perspective rather than a collection of development projects
- Build tools to translate business strategy into a portfolio of innovation initiatives, and to measure performance of uncertain and long-term innovation
- Build processes and structures to enhance the capability of organization to generate creative ideas for different purposes and with different degrees of novelty

References

[1] Roussel, Philip A. Third generation R&D: managing the link to corporate strategy. Harvard Business Press, 1991.

[2] Howells, John. The management of innovation and technology: the shaping of technology and institutions of the market economy. Sage, 2005.

ii. Product Design (3 Credits)

Contents

Mechanical design, thermal design, industrial design, modelling of product metrics, design for manufacture and assembly, design for environment, design for reuse, design for reliability, design for six sigma, design for environment, modular design, design for robustness, analytical and numerical model solutions

Objectives

- provide an integrated knowledge and skills in product/industrial design;
- develop design and technology research skills related to the design process through practicing applied research;
- develop advanced design skills, enabling graduates to practice as an independent design professional and to further develop design and professional skills in product/industrial design engineering;
- nurture scientific rigour as well as creativity to enable graduates to follow a successful career in product/industrial design and assume leadership roles in national and international companies and institutions.

References

[1] Chitale, A. K. Product design and manufacturing. PHI Learning Pvt. Ltd., 2011.

[2] Mascitelli, Ronald. The lean product development guidebook: everything your design team needs to improve efficiency and slash time-to-market. Technology Perspectives, 2007.

[3] Ward, Allen C., and Durward K. Sobek II. Lean product and process development. Lean Enterprise Institute, 2014.

[4] El-Haik, Kai Yang-Basem, and K. Yang. "Design for Six Sigma, A Roadmap for Product Development." (2003): 1-35.

iii.	**Product Development (3 Credits)**

Contents

An overview of successful product development, critical scientific thinking, approaches of product development and project management, product design: structure and process, consideration of materials, selection of manufacturing processes and design considerations, designing for maintenance, consideration of usability, designing for functionality, product cost consideration, CAD/CAE/VR/PLM tools to support the development of a product, product applications and manufacturing processes, sustainability issues in product development, lean, agility and QRM philosophies in product development

Objectives

- Develop knowledge, critical scientific thinking and hands-on experiences for developing a product
- Demonstrate a scholarly approach of product development and project management and evolution and use of the most suitable material and technology
- Demonstrate ability to carry out research into customer and market requirements and their analysis to translate the requirements into product specification
- Demonstrate knowledge of selection of materials and manufacturing processes
- Demonstrate skill of using CAD/CAE/VR/PLM tools to support the development of a product
- Apply materials appropriately to product applications and manufacturing processes
- Demonstrate a knowledge of Advanced Manufacturing Processes and Rapid Prototyping techniques
- Demonstrate knowledge of considerations of sustainability issues in product development.

References

[1] Ulrich, K. T., Eppinger, S. D., &Goyal, A. (2011). *Product design and development* (Vol. 2). New York: McGraw-Hill.

[2] Annacchino, M. A. (2003). *New product development: from initial idea to product management*. Butterworth-Heinemann.

iv. Advanced prototyping (3 Credits)

Contents

Advanced Prototyping is practical oriented course containing physical prototyping and virtual prototyping, physical and virtual testing (CFD Analysis) of prototypes, analysis with focus on employing the latest rapid prototyping (RP) technologies, and providing a wide range of creative resources incorporating traditional techniques, computer-aided design and computer-aided manufacturing.

Objectives

- To pursue developmental research that examines new rapid prototyping techniques and processes, scanning, selective laser sintering and 3D printing while exploring new and modified materials for manufacture
- To investigate design for manufacture, sustainable design and strategic design concepts and prototypes with the potential of providing significant value to industry.

References

[1] Noorani, Rafiq. Rapid prototyping: principles and applications. John Wiley & Sons Incorporated, 2006.

[2] Venuvinod, Patri K., and Weiyin Ma. Rapid prototyping: laser-based and other technologies. Springer Science & Business Media, 2003.

[3] Gibson, Ian, David W. Rosen, and Brent Stucker. Additive manufacturing technologies. New York: Springer, 2010.

[4] Hopkinson, Neil, Richard Hague, and Philip Dickens, eds. Rapid manufacturing: an industrial revolution for the digital age. John Wiley & Sons, 2006

v.	**Advanced Manufacturing Processes (3 Credits)**

Contents

Mechanical behavior of metals, structure and properties of metals, casting processes (sand casting analysis), forging and extrusion analysis, rolling and drawing analysis, sheet metal (shearing/bending/stretch/spinning) analysis, composite manufacturing (particle/fibre reinforced), shaping process of plastics, metal machining (orthogonal cutting, force calculation, merchant mode, turning, drilling, milling and grinding analysis), CNC machining, and nonconventional machining (EDM, ECM, Laser beam, ultrasonic, abrasive jet and water jet)

Objectives

- Provide students with an understanding of the latest technology being used in manufacturing industries in the area of Computer Aided Design and Computer Aided Manufacturing.

- Allow students to use the CNC machines in the lab for the manufacture of parts, which have been designed on the CAD packages.
- Understand and appreciate the importance of basic principles of Manufacturing Systems.
- Understand the application of those principles in practice

References

[1] Kalpakjian, Manufacturing Processes for Engineering Materials, Wesley, 4th Ed.

[2] E. Paul DeGarmo, Materials and Processes in Manufacturing, Prentice Hall, 8th Ed., 1997.

[3] Mikell P. Groover, Fundamentals of Modern Manufacturing, 2nd Ed, Prentice-Hall, 2002.

vi.	**Product Quality and Assurance (3 Credits)**

Contents

Introduction to metrology and quality control, general principles of measurements, dimensioning and geometrical tolerances, comparators/surface texture measurement, statistical process control, control charts for variables I and II, fundamental of probability, control charts for attributes I and II, lot by lot acceptance sampling for attributes, acceptance sampling plan, TQM tools, techniques and principles, Six sigma philosophies.

Objectives

- recognize how inputs to Perform Quality Assurance are used in the process
- recognize examples of activities that are conducted in an effective quality audit
- recognize examples of issues that may trigger the need for a process analysis
- identify the outputs of performing quality assurance
- differentiate between attribute and variables sampling
- identify tolerances and control limits, given a sample quality control graph

References

[1] Montgomery, D. C. (2013). Applied Statistics and Probability for Engineers 6th edition. Wiley.

[2] Vasconcellos, J. A. (2003). Quality assurance for the food industry: a practical approach. CRC Press.

vii. Product standardizations (3 Credits)

Contents

Product quality standards, material standards (Material selection, qualification and testimg), manufacturing, safety and qualification standards, environmental standards, fits and tolerance standards.

Objectives

- An ability to understand the industrial standards and apply them in real mode

References

[1] Guasch, J. Luis. Quality systems and standards for a competitive edge. World Bank Publications, 2007.

[2] Kverneland, Knut O., and Knot O. Kverneland. Metric Standards for Worldwide Manufacturing. ASME Press, 2007.

[3] Pelton, Joseph N., and Ram Jakhu, eds. Space Safety Regulations and Standards. Elsevier, 2010.

viii. Marketing and product commercialization (3 Credits)

Contents

Market-oriented strategic planning, marketing research and information systems, buyer behavior, target market selection, competitive positioning, product and service planning and management, pricing, distribution, and integrated communications, including advertising, public relations, Internet marketing, social media, direct marketing, and sales promotions.

Objectives

- An ability to make effective marketing decisions, including assessing marketing opportunities and developing marketing strategies and implementation plans

References

[1] Rafinejad, Dariush. Innovation, Product Development and Commercialization: Case Studies and Key Practices for Market Leadership. J. Ross Publishing, 2007.